

LTV / ROAS

Prediction Engine

VERSION 3.39

March 17, 2026 · Comprehensive Technical Documentation

17

Prediction
Horizons

D1-D390

Forecast
Range

2-Layer

Model
Architecture

700+

Calibration
Cells

Casual Title

**Predictive
Analytics**

Executive Summary

LTV / ROAS Prediction Engine v3.39

The Casual Title LTV/ROAS Prediction Engine predicts Lifetime Value and Return on Ad Spend across 17 horizons (D1-D390) for mobile game cohorts, enabling data-driven budget allocation and campaign optimization.

Data Source

Raw cohort-level CSV exports from Adjust MMP, containing installs, cost, LTV at fixed horizons (D0-D120 + M4-M12), and retained user counts.

Core Architecture

Two-layer model: Layer 1 fits power-law reference curves per OS × Geo × Channel. Layer 2 computes per-cohort scaling factors (SF) with 6 post-processing steps.

v3.39 Key Changes

D30-D90 input accuracy via input-day-aware SF decay, niche channel handling, early-input confidence discount, Aura from Unity support, parallel backtest execution.

OFFICIAL METRIC

Install-weighted MAPE

INPUT WINDOW

D1 / D3 / D7 / D14 / D30 / D60 / D90 / D120

CONFIDENCE LEVELS

High / Medium / Low (4-component score)

System Architecture & Data Flow

01

CSV Ingestion

load_adjust_data()
Clean, standardize channels,
enforce LTV monotonicity,
assign geo brackets

02

Reshape

reshape_long()
Convert wide cohort data →
long format (one row per
cohort × horizon)

03

Reference Curves

build_reference_curves()
Fit power-law curves per OS ×
GeoKey × Channel (Layer 1)

04

Retention & Tail

build_retention_reference() +
build_tail_reference()
Compute retention & LTV ratio
references

05

SF Prior Lookup

build_sf_prior_lookup()
Build historical SF lookup for
future cohorts with no
observed data

06

Predictions

generate_predictions()
Main loop: per-cohort LTV at all
17 horizons with 6 post-
processing steps

07

Backtest [opt]

run_backtest()
Simulate historical predictions
output accuracy matrices
(parallel)

Two-Layer Model

Layer 1: Power-law reference
curves

Layer 2: Per-cohort scaling
factors

Two-Layer Prediction Model

How each cohort LTV forecast is generated

LAYER 1 · Reference Curves

$$\text{LTV}(d) = a \times d^b$$

- Power-law model fitted per OS × GeoKey × Channel
- Fitted only through D120 (avoids vintage bias)
- Install-weighted; OS-specific recency weighting
- Fallback chain: country → geo-bracket → OS-only
- Vintage-bias recovery for AppLovin & similar



LAYER 2 · Scaling Factors (SF)

$$\text{SF} = \frac{\text{observed_LTV}}{\text{reference_LTV}}$$

- Weighted median across observed horizons (up to D120)
- Captures how much a cohort deviates from segment average
- Clipped per OS and channel; relaxed for early inputs
- SF dampening for D>120 tail (log-space decay toward target)
- SF prior lookup when cohort has no observed data yet

Post-Processing Pipeline

6 refinement layers applied after the base Layer 1 × Layer 2 prediction

1

Retention Adjustment

$$ret_ratio^{(\alpha \times \ln(h/d, 1))}$$

Adjusts for cohort retention vs reference. Alpha=0.5 default; channel-specific overrides. Tail-anchor boost for very early inputs at D300-D390.

2

Bridge Shape Correction

$$correction = empirical_ratio / curve_ratio$$

For Mintegral, Unity, Google at early inputs. Replaces curve's D14→D120 growth ratio with empirical data. Clipped [0.2, 5.0].

3

Tail Factors (D>120)

$$tail = LTV(h) / LTV(D120)$$

Install-weighted median ratio from historical data. Clipped [1.0, 3.0], enforced monotone. Vintage-aware buckets for drift-prone channels.

4

Campaign-Level Tail Cal.

$$shrunk\ correction\ per\ campaign$$

Shrunk campaign-specific tail correction on top of vintage tail factor at D150-D300 (iOS/Android Mintegral).

5

Residual Calibration

$$cal_factor = 1 + strength \times (1/prior - 1)$$

3 prior tables: Segment-level (72+ cells), Geo-aware (700+ cells), Incent fallback (30 cells). Fixes systematic bias.

6

Confidence Discount

$$blend = (h - D120) / (D390 - D120) \times 0.25$$

For D1-D3 inputs predicting D>120: blends SF prediction toward reference (SF=1). 0% blend at D120, 25% blend at D390.

Residual Calibration System

Three-tier lookup that fixes systematic bias not captured by the base model

$$\text{cal_factor} = 1.0 + \text{strength} \times (1 / \text{prior_ratio} - 1.0) \quad [\text{clipped per tier}]$$

TIER 1 · Segment-Level Prior

Lookup Key

(os, channel, input_day_bucket, horizon_bucket)

CELLS

72+ cells

STRENGTH

0.50

CLIP

(0.65, 1.12)

Covers all major OS × Channel combos.
Input day buckets: 1, 14, 30, 60, 90.
Horizons: near (≤D120), mid (D150-D240), far (D270+).

TIER 2 · Geo-Aware Prior

Lookup Key

(os, channel, geo_cal_group, input_day_bucket, horizon_bucket)

CELLS

700+ cells

STRENGTH

0.80

CLIP

(0.28, 1.25)

Targets ROW bias (historically 2-4.5× overprediction). Geo groups: us, tier1_asia, row_prem, row_other.

TIER 3 · Incent Fallback Prior

Lookup Key

(os, input_day_bucket, horizon_bucket)

CELLS

30 cells

STRENGTH

0.50

CLIP

(0.65, 1.12)

Applies to all incent channels as a group when channel-specific priors are unavailable.

What's New in v3.39

Fix 3.1

Input-Day-Aware SF Decay

For D30-D90 inputs, SF decay is now anchored to the latest observed day rather than averaging all observed horizons. Prevents over-smoothing for mid-window inputs and improves accuracy on recently acquired cohorts.

Fix 3.2

Niche Channel Handling

Channels with <1,000 total installs get wider SF clipping (0.001-10.0) and reduced residual calibration strength (0.20). Prevents hard-floor forcing and overfitting of sparse priors.

Fix 3.3

Early-Input Confidence Discount

D1-D3 inputs predicting D>120 are blended toward the reference curve (SF=1): 0% at D120, 25% at D390. Shrinks extreme extrapolations from the earliest, most uncertain input window.

Fix 3.4a

Wider Residual Clip for D1-D3

Residual calibration clip for bucket=1 expanded from (0.65, 1.12) to (0.50, 1.25), allowing stronger bias correction for the most uncertain input windows.

Fix 3.5

Aura from Unity Channel

"Aura from Unity" added as a first-class channel with proper standardization and treatment as an incant channel. Previously misclassified or unhandled in earlier versions.

Fix 3.6

Parallel Backtest Execution

Backtest passes now run in parallel across all CPU cores using multiprocessing.Pool with fork start method. Shared objects inherited via copy-on-write — zero extra memory overhead.

Geographic & Channel Classification

GEOGRAPHIC CLASSIFICATION

Tier 1 US, GB, AU, CA, DE, FR, ...

Tier 1 Asia JP, KR, TW, HK, SG

ROW Premium Select emerging markets

ROW Other All remaining countries

Fallback Chain: Country → Geo-Bracket → OS-Only

CHANNEL CLASSIFICATION

Non-Incent Channels

Mintegral

Google

AppLovin

Unity

Apple

Meta

TikTok

Incent Channels (examples)

Gamelight

Adjo

Tapjoy

Tyrad

Torox

Prodege

Digital Turbine

Aura from Unity (new)

Backtest System & Accuracy

Rigorous retrospective validation across all input windows

Methodology

For each input day $D \in \{1, 3, 7, 14, 30, 60, 90, 120\}$:

- Mask each cohort to only use LTV observations up to day D
- Run full prediction pipeline for all horizons beyond D
- Compute $APE = |\text{predicted} - \text{actual}| / \text{actual}$ per cohort
- Aggregate as install-weighted MAPE across cohorts

Output Files

- accuracy_breakdown.xlsx — per-segment MAPE
- accuracy_matrix_mape.csv — full matrix
- accuracy_matrix_within15.csv — % within $\pm 15\%$
- accuracy_matrix_n.csv — cohort counts

Parallel Execution (v3.39)

multiprocessing.Pool with 'fork' start method · Shared DataFrames and reference objects inherited via copy-on-write · Zero extra memory overhead · Scales with available CPU cores

Official Metric

Always use install-weighted MAPE — not revenue-weighted. Revenue-weighted diverges for Android ROW where high-install / near-zero-LTV cohorts dominate install weights but high-LTV US cohorts dominate revenue.

Known Pitfalls & Best Practices



"tier1" Keys Are Dead

Always use "tier1_asia". Any prior keyed with "tier1" will never be matched. The engine asserts this at startup.



Back Out Existing Calibration

When computing new priors:
$$\text{uncal_bias} = \text{predicted_ltv} / \text{residual_calibration_factor} / \text{actual_ltv}.$$

Failing to do this double-counts correction.



MAPE Divergence (Android ROW)

Install-weighted vs revenue-weighted MAPE can show opposite bias directions. Always use the official install-weighted metric.



Adjust LTV Back-Corrections

load_adjust_data() applies cummax() to enforce LTV monotonicity. Old CSV files without this step may show decreasing LTV values.



Ref Curves Fit Only Through D120

Do not raise REF_FIT_MAX_DAY beyond 120 without validation. Long-horizon fitting introduces vintage bias from older cohorts dominating D300+ data.



iOS Recency Weighting Off

iOS monetization is stable across vintages. Enabling recency weighting for iOS causes systematic underprediction. Keep it disabled.

LTV / ROAS Engine

v3.39 · Casual Title

Two-layer power-law model with 6 post-processing refinement steps

700+ geo-aware residual calibration cells for precision bias correction

Rigorous parallel backtest across 8 input windows with install-weighted MAPE

v3.39 delivers improved D30-D90 accuracy, niche channel safety & confidence discounting